

Database Systems Development

M30232

CW Group contribution statement

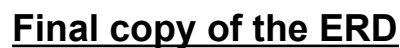
Group 12

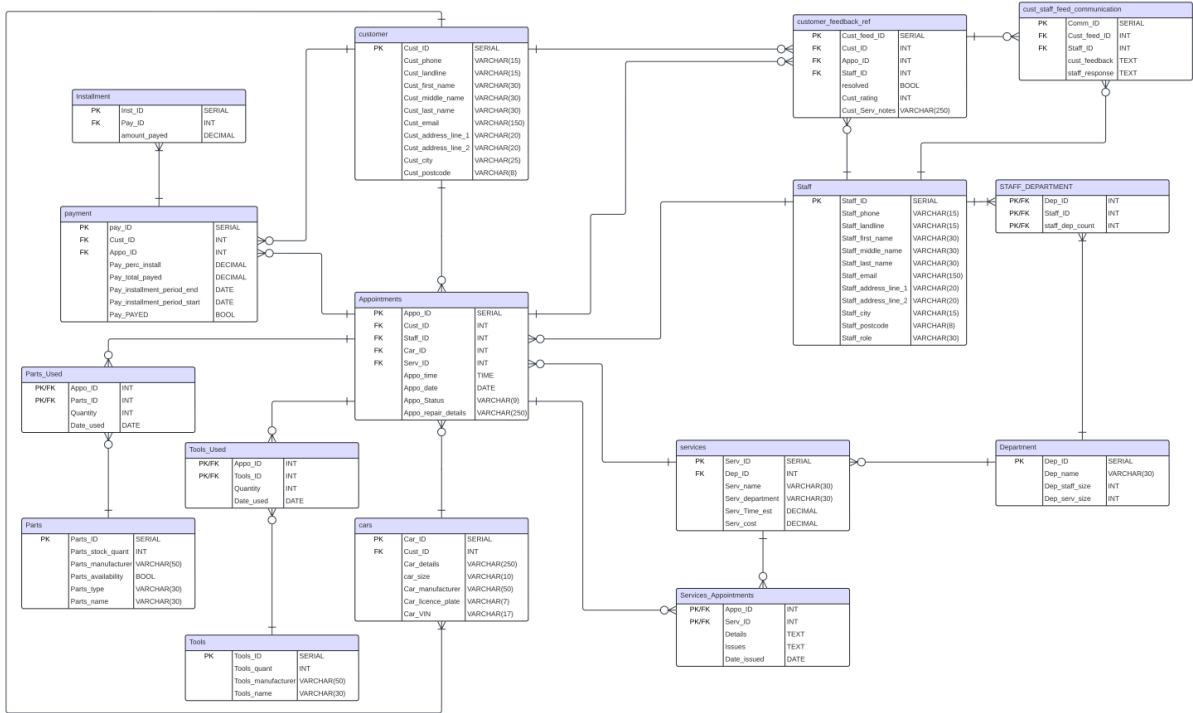
	UP2274400	UP2299667	UP2301043
ERD Design	☐	☒	☒
Reflective Analysis	☒	☐	☐
Data Dictionary	☒	☒	☒
Document writing	☐	☒	☐
Brainstorming	☒	☒	☒
SQL Coding	☒	☐	☒
DDT	☒	☒	☒
Percentage Allocation	33.3%	33.3%	33.3%
Signature	Finn cullen	Isaac Lawal	Ethan White

Contribution Description

UP NUMBER	Description (if not equally counted)	Signature
UP2274400		Finn Cullen
UP2299667		Isaac lawal
UP2301043		Ethan White

UP2274400s Initial ERD





CUSTOMER						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Cust_ID	PK		SERIAL			
Cust_phone	AK	Y	VARCHAR(15)	NOT NULL, UNIQUE		
Cust_landline			VARCHAR(15)	NOT NULL		
Cust_first_name		Y	VARCHAR(30)	NOT NULL		
Cust_middle_name			VARCHAR(30)			
Cust_last_name		Y	VARCHAR(30)	NOT NULL		
Cust_email	AK	Y	VARCHAR(150)	NOT NULL, UNIQUE		
Cust_address_line_1			VARCHAR(20)	NOT NULL		
Cust_address_line_2			VARCHAR(20)			
Cust_city			VARCHAR(25)	NOT NULL		
Cust_postcode		Y	VARCHAR(8)	NOT NULL		

STAFF						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Staff_ID	PK		SERIAL			
Staff_phone	AK	Y	VARCHAR(15)	NOT NULL, UNIQUE		
Staff_landline			VARCHAR(15)	NOT NULL		
Staff_first_name		Y	VARCHAR(30)	NOT NULL		
Staff_middle_name			VARCHAR(30)			
Staff_last_name		Y	VARCHAR(30)	NOT NULL		
Staff_email	AK	Y	VARCHAR(150)	NOT NULL, UNIQUE		
Staff_address_line_1			VARCHAR(20)	NOT NULL		

Staff_address_line_2			VARCHAR(20)			
Staff_city			VARCHAR(25)	NOT NULL		
Staff_postcode		Y	VARCHAR(8)	NOT NULL		
Staff_role			VARCHAR(30)	NOT NULL		What role the staff member has.

DEPARTMENT						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Dep_ID	PK		SERIAL			
Dep_name			VARCHAR(30)	NOT NULL		
Dep_staff_size			INT	NOT NULL, CHECK (Dep_staff_size >= 0)		How many staff per department
Dep_serv_size			INT	NOT NULL, CHECK (Dep_serv_size >= 0)		How many services per department

STAFF_DEPARTMENT						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Dep_ID	PK/FK		INT	NOT NULL	DEPARTMENT. Dep_ID	
Staff_ID	PK/FK		INT	NOT NULL	STAFF.Staff_ID	
staff_dep_count			INT	NOT NULL		Used to keep each row unique by incrementing every time a staff ID repeats.

PARTS						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
parts_ID	PK		SERIAL			
parts_stock_quant		Y	INT	NOT NULL, CHECK(parts_stock_quant >= 0)		How many parts we have

parts_manufacturer			VARCHAR(50)	NOT NULL		
parts_availability		Y	BOOLEAN	NOT NULL		Is it currently available?
parts_type			VARCHAR(30)	NOT NULL		What aspect of the car will it be used in (I.E engine, suspension, electrical)
parts_name		Y	VARCHAR(30)	NOT NULL		

TOOLS						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
tools_ID	PK		SERIAL			
tools_quant		Y	INT	NOT NULL, CHECK(tools_quant >= 0)		How many tools are in stock
tools_manufacturer			VARCHAR(50)	NOT NULL		
tools_name		Y	VARCHAR(30)	NOT NULL		

CARS						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Car_ID	PK		SERIAL			
Cust_ID	FK		INT	NOT NULL	CUSTOMER.Cust_ID	
Car_details			VARCHAR(250)	NOT NULL		Space for any details such as condition or problems
Car_size			VARCHAR(10)	NOT NULL		Denotes what kind of car it is and size such as truck,medium,small,etc
Car_manufacturer			VARCHAR(50)	NOT NULL		
Car_licence_plate	AK	Y	VARCHAR(7)	NOT NULL, UNIQUE		
Car_VIN	AK	Y	VARCHAR(17)	NOT NULL, UNIQUE		

SERVICES

<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Serv_ID	PK		SERIAL			
Dep_ID	FK		INT	NOT NULL	DEPARTMENT.Dep_ID	
Serv_name			VARCHAR(30)			
Serv_department		Y	VARCHAR(30)			Associated department of the service
Serv_Time_est			DECIMAL	NOT NULL, CHECK (Serv_Time_est >= 0)		Estimation of time taken
Serv_cost		Y	DECIMAL	NOT NULL, CHECK (Serv_cost >= 0)		Estimation of cost

APPOINTMENT

<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Appo_ID	PK		SERIAL			
Cust_ID	FK		INT	NOT NULL	CUSTOMER.Cust_ID	
Staff_ID	FK		INT	NOT NULL	STAFF_Staff_ID	
Car_ID	FK		INT	NOT NULL	CARS.Car_ID	
Serv_ID	FK		INT	NOT NULL	SERVICES.Serv_ID	
Appo_time		Y	TIME	NOT NULL		What time it happens
Appo_date		Y	DATE	NOT NULL		What date it happens
Appo_Status		Y	VARCHAR(9)	NOT NULL		Current status (I.E in

						progress,completed,on hold)
Appo_repair_details			VARCHAR(250)			Room for special details about the appointment

SERVICES_APPOINTMENTS						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Appo_ID	PK/FK		INT	NOT NULL	APPOINTMENTS.Appo_ID	
Serv_ID	PK/FK		INT	NOT NULL	SERVICES.Serv_ID	
Details			TEXT	NOT NULL		Room for details
Issues			TEXT			Room for issues encountered
Date_issued			DATE	NOT NULL		

CUSTOMER_FEEDBACK_REF						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Cust_feed_ID	PK		SERIAL			
Cust_ID	FK		INT	NOT NULL	CUSTOMER.Cust_ID	
Appo_ID	FK		INT	NOT NULL	APPOINTMENTS.Appo_ID	
Staff_ID	FK		INT	NOT NULL	STAFF.Staff_ID	
resolved			BOOLEAN	NOT NULL		Is the customer feedback completed
Cust_rating			INT	CHECK (Cust_feed_rating >= 0)		A 0-5 star rating system
Cust_Serv_notes			VARCHAR(250)			Room for notes on the exchange

CUST_STAFF_FEED_COMMUNICATION						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Comm_ID	PK		SERIAL			
Cust_feed_ID	FK		INT	NOT NULL	CUSTOMER_FEEDBACK_REF.Cust_feed_ID	
Staff_ID	FK		INT	NOT NULL	STAFF.Staff_ID	
Cust_comm_feed back			TEXT			
Cust_comm_responce			TEXT			

PARTS_USED						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Appo_ID	PK/FK		INT	NOT NULL	APPOINTMENTS.Appo_ID	
parts_ID	PK/FK		INT	NOT NULL	PARTS.parts_ID	
Quantity			INT	NOT NULL, CHECK (Quantity > 0)		
Date_used			DATE	NOT NULL		

TOOLS_USED						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
Appo_ID	PK/FK		INT	NOT NULL	APPOINTMENTS.Appo_ID	

tools_ID	PK/FK		INT	NOT NULL	TOOLS.tools_ID	
Quantity			INT	NOT NULL, CHECK (Quantity > 0)		
Date_used			DATE	NOT NULL		

PAYMENT						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
pay_ID	PK		SERIAL			
Cust_ID	FK		INT	NOT NULL	CUSTOMER.Cust_ID	
Appo_ID	FK		INT	NOT NULL	APPOINTMENTS.Appo_ID	
Pay_perc_install		Y	DECIMAL	NOT NULL, CHECK (Pay_perc_install > 0)		Percentage paid per installment, measured as a decimal so 50.6 = 50.6% per installment
Pay_total_payed			DECIMAL	NOT NULL, CHECK (pay_total_payed > 0)		Total cost
Pay_installment_period_end			DATE	CHECK (Pay_installment_period_end >= Pay_installment_period_start)		When they finished paying
Pay_installment_period_start			DATE	NOT NULL		When they started paying
Pay_PAYED		Y	BOOLEAN	NOT NULL		Has it been payed

INSTALLMENT						
<i>Attribute_Name</i>	<i>KEY</i>	<i>INDEX</i>	<i>Data Type & Size</i>	<i>Domains & Constraints</i>	<i>FK Reference</i>	<i>Description</i>
inst_ID	PK		SERIAL			
Pay_ID	FK		INT	NOT NULL	PAYMENT.P	

					ay_ID	
amount_payed		Y	DECIMAL	NOT NULL, CHECK (amount_paye d > 0)		How much was paid this installment

Reflective analysis

Due to 2274400's ERD design having received a substantially high mark in cw1, few changes were made to the design with some improvements taken from both feedback and the database design provided by 2299667 and 2301043.

Firstly the tools and tools_used were added from 2299667 and 2301043's database design to account for any tools that would have been used in the servicing of the vehicles, keeping track of these tools would have been imperative for the company.

Secondly revisions were made to the design by 2299667 and 2301043 who added the staff_ID foreign key to the revised cust_staff_feed_communications table to reflect the relationship between staff and cust_staff_feed_communications tables with staff having many cust_staff_feed_communications.

Thirdly the service_report table has been changed to the service_appointments intersection table where it retains its ability to hold service reports for the appointment table but has also been repurposed into a intersection table between service and appointment so that a many to many relationship could be eliminated that was present in 2274400's design.

Fourthly indexing has been added to the database as seen in the data dictionary where data that would be frequently queried such as phone numbers, part quantities or appointment statuses has been indexed to improve the efficiency of querying.

Lastly several changes to data types and logic were made including changing some CHECK values in the data dictionary to include or exclude certain values that shouldn't have been allowed such as the start date being later than the end date or allowing a 0 star rating.

Additionally appo_status was changed to varchar(9) as it didn't need to be TEXT, quantity within parts_used and tools_sued is no longer a key and Rep_ID has been removed, these changes have all been made to optimise the database by reducing the storage reserved for certain attributes or removing attributes that are considered unnecessary.

Database Systems Development M30232

Group Number: [12]

Database Development Tracker

Date	Task Description	Member ID	Task Details	Time	Signature
21/01/2025	SQL coding : making the tables	UP2274400	I implemented half of the database using my ERD as a template, with this the customer,staff,department,staff_department,parts,cars and services tables were successfully implemented with full constraints, keys and data types.	1H	Finn Cullen
23/01/2025	SQL coding : finishing the tables	UP2274400	I finished implementing the tables this day adding the appointment,service_report,customer_feedback_ref,cust_feed_communication,parts_used,payment and installment tables, with this the first iteration of the database is completed and should be in such a state that when revisions and improvements are made they can swiftly be implemented.	1H	Finn Cullen
25/01/2025	SQL coding : testing	UP2274400	The first iteration of the table was tested, there were some missing commas and incorrect syntax but nothing ultimately difficult, additionally the tables were reordered in the SQL file to ensure that when they are mass inserted then they all function properly.	0.5H	Finn Cullen
31/01/2025	Initial meeting	UP2274400 UP2299667	We have met initially to discuss the CW details, hand out roles and organise	1H	Ethan White Finn Cullen

		UP2301043	future meetings and forms of communication.		
02/02/25	Initial ERD editing/notes	UP2301043	Making notes for any potential changes and implementing some. Including cust_staff_feed_communication, Tools, Tools_Used, Service_Appointments	1H	Ethan White
03/02/25	SQL coding : Inserts	UP2274400	Here we begin implementing the inserts for the database, with customers and staff being completed, mockaroo was used to generate mass data for the databases.	1H	Finn Cullen
04/02/25	ERD editing	UP2301043 UP2299667	We looked at the feedback for both ERDs and made some edits to the new ERD incorporating elements from both ERDs and altering some names and connections based on the feedback. Finalising the changes made in the initial ERD editing. Tools, Tools_Used fully added and changes to attributes in Services_Appointments.	0.5H	Ethan White
04/02/25	SQL coding : Inserts	UP2274400	Today the department and staff department were completed, both were reasonably short and easy to implement causing no notable issues.	0.5H	Finn Cullen
04/02/25	SQL coding : Inserts	UP2274400	Today parts and cars were completed which are among the largest database inserts to be completed, unfortunately due to the unique information to be inserted (manufacturer, cost, VIN, ECT) programs such as mockaroo can no longer be relied on for inserts causing my workload to balloon. No notable issues to report	1H	Finn Cullen
04/02/25	Meetup for catching up	UP2274400 UP2299667 UP2301043	Another meetup so we can check everything is in order. Showing the code that 2274400 had done to this point and the ERD changes made.	0.5H	Ethan White Finn Cullen
06/02/25	SQL coding : Inserts	UP2274400	Today services and appointments were completed, appointments posed a considerable issue as it has 4 foreign keys, additionally i don't know how DATE and TIME is to be handled in SQL so i will ask in my next DSD lab for guidance and later change them to the correct format	1H	Finn Cullen
07/02/25	SQL coding : Inserts	UP2274400	Service report and customer_feedback_ref are completed,	1H	Finn Cullen

			similar to appointments these require a lot of foreign keys which is a chore and services such as mockaroo have become all but useless for they cannot generate the data required, i have additionally realised that TRUE and FALSE don't work for SQL as boolean inserts so i will add it to the list of changes to be retroactively made		
08/02/25	SQL coding : Inserts	UP2274400	Customer feedback communication and parts used are completed, for the comms table I gave up on making proper communication and have used lorem ipsum to fill the table, additionally parts used were difficult as I had to pull data from several tables.	1H	Finn Cullen
09/02/25	SQL coding : Inserts	UP2274400	Inserts are finally complete with Payment and installment, i have settled for only allowing 2 payment plans so i don't get a migraine, 100% upfront payment and 50% 2 time payment, with tis now all that needs to be done is to fix the data types and test the inserts	1H	Finn Cullen
09/02/25	Putting the documentation together	UP2301043 UP2274400 UP2299667	Using the templated to insert the DDT entries and the contribution statement	0.5H	Ethan White
11/01/25	SQL coding : Inserts and debugging	UP2274400	All boolean, TIME and DATE inserts were fixed to be in the correct format and the inserts were all tested for errors, other than having to change 100+ DATE inputs there were only a few errors within the inserts so other than tedious DATE replacements the development went smoothly. With this the database V1.0 is finalised and only needs some updating with the finalised ERD design to be truly completed.	1H	Finn cullen
11/02/25	Data Dictionary entries	UP2301043 UP2299667	Using the ERD completed the Data dictionary excluding INDEX-es, descriptions and Alternate keys. All to be added later.	1.25H	Ethan White
15/02/25	SQL coding : final revisions	UP2274400 Up2299667	With this the database is all but completed, the new tables of TOOL and TOOLS_USED are implemented, changes regarding the feedback given	1H	Finn Cullen

			such as SERVICE_REPORT becoming SERVICE_APPOINTMENT as an intersection table alongside CUST_FEED_COMMUNICATION gaining the staff_ID foreign key. Additionally I added the indexes for the database, these indexes cover frequently queried data such as phone numbers or the status of appointments.		
15/02/25	Finalising the Data Dictionary	UP2301043 UP2274400 UP2299667	Adding the INDEX-es that were implemented in code form. Adding any possible AK and making sure any code changes were represented. Descriptions added by 2274400	0.5H	Ethan White
17/02/2025	Reflective analysis	up2274400	I compiled a list of the changes made during cw2, within the analysis i went through the list and grouped similar changes such as data type changes or table name changes, i then went through with explaining why each change was made.	0.5H	Finn Cullen